

Cloud Agility

- Many of the same methods used in **agile software development** are now being used in **cloud computing**.
- These methods help teams work in a **flexible, adaptive, and faster way** when managing cloud systems.

Cloud agility means using *agile development ideas* (like flexibility and fast updates) in *cloud environments* to make systems faster, easier to change, and more responsive.

Project Management



- A **project** includes many processes grouped into **five stages**: **initiating, planning, executing, monitoring & controlling, and closing**.
- **Project management** means using **specific skills, tools, and techniques** to manage these processes so that the **desired goal** is successfully achieved.
- Important **project management skills** include:
 - **Communication** – Sharing clear inf. between team members.
 - **Negotiation** – Reaching agreements and solving conflicts.
 - **Task and time management** – Organizing work and meeting deadlines.
 - **Cost and quality management** – Controlling budget and maintaining quality.
 - **Risk management and leadership** – Predicting problems and guiding the team.

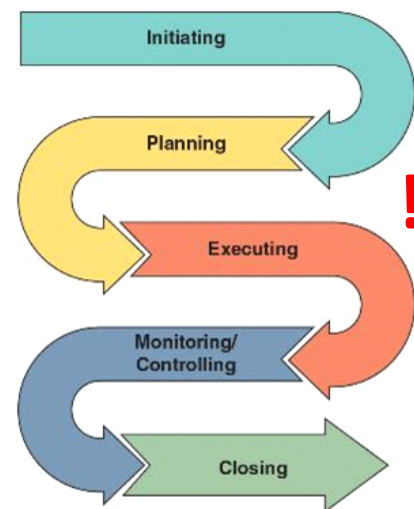


Figure 3-7 The five project management process groups

Project management is about planning, organizing, and controlling all parts of a project — using communication, time, cost, and leadership skills to reach goals effectively.

ALM (application lifecycle management)

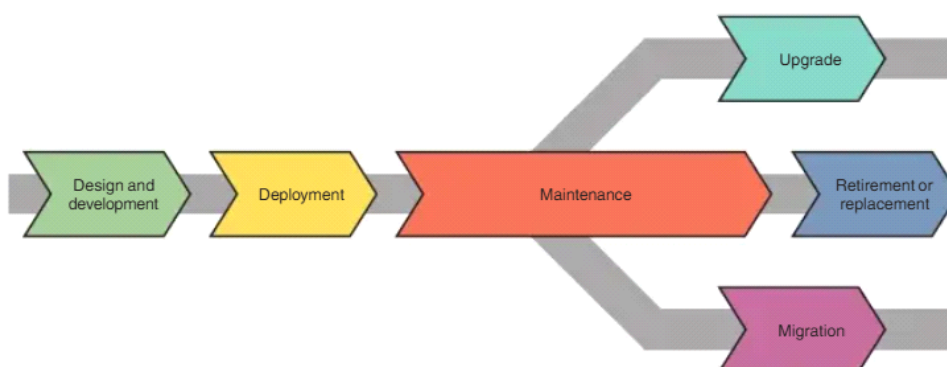


Figure 3-8 ALM (application lifecycle management) phases

- **Upgrade** – The app is improved or gets new features.
- **Migration** – The app is moved to a new environment, such as the cloud.
- **Retirement or Replacement** – The app is removed or replaced by a new one.

ALM helps manage the whole life of an application — from the first design to its upgrade or replacement — keeping it useful and efficient.

Application Life Cycle – Waterfall Method

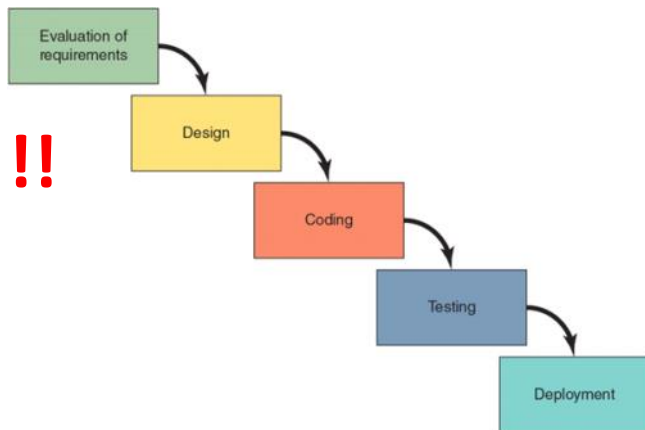


Figure 3-9 Waterfall method

- In the past, software development followed a **linear process** called the **Waterfall Method**.
- Each step had to be **finished completely** before moving to the next one.

! Application Life Cycle – Cyclical Approach

- The **cyclical (repeating) model** was created to make software development **faster and more flexible** than the old linear waterfall method.
- It allows teams to **adapt quickly** to market or business changes.
- Each cycle helps improve the software continuously.

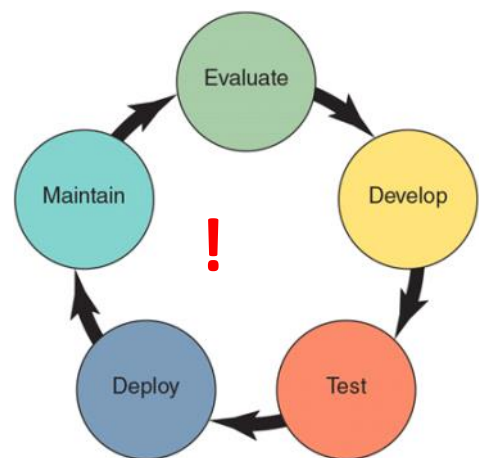


Figure 3-10 A cyclical approach to software development

Application Life Cycle – DevOps

- The modern version of the application life cycle is **DevOps**, which combines **Development** and **Operations**.
- DevOps makes the process **continuous, faster, and more collaborative**.
- Teams plan, build, test, deploy, and monitor applications in a **repeating loop**, ensuring constant improvement.

Main Steps in DevOps Cycle:

1. **Plan** – Define goals and what to build.
2. **Build** – Develop the code or new features.
3. **Continuous Integration** – Combine and test new code often.
4. **Deploy** – Release updates quickly and safely.
5. **Operate** – Keep the system running smoothly.
6. **Continuous Feedback** – Use feedback to improve the next cycle.

- DevOps helps teams work **faster**, reduce errors, and **adapt quickly** to new business needs.

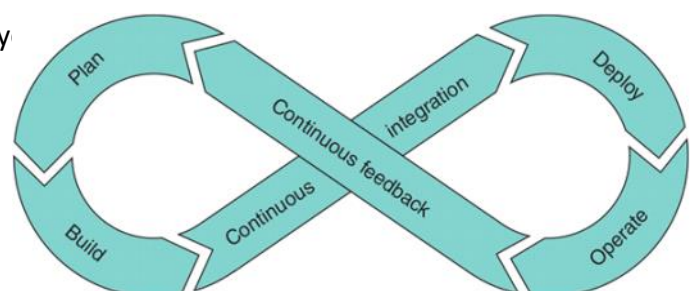


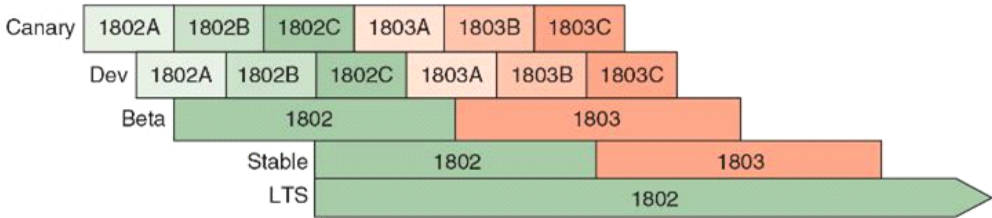
Figure 3-11 DevOps blends development and operations functions

Builds

- A **build** means creating a **working version** of an app, website, or system from the **source code files**.
- In **DevOps**, using **build automation** helps teams release updates **faster** and with **less manual work** — improving agility.
- Repeated versions or updates of the same build are called a **channel** (for example: beta channel, stable channel).

Short meaning:

A build is the **ready-to-use version** of your code.
Automation makes builds quicker, and each version belongs to a channel.



Channel	Who Uses It	What It Means
Canary 🐣	Internal testers	The very first, risky version. New code gets tested here. It often crashes.
Dev 🧑💻	Developers	A bit more stable, used to fix early bugs before public testing.
Beta 🟦	Beta users (public testers)	Almost ready, but still may have small issues. Public users can test it.
Stable ✅	Everyone	The official version you download — fully tested and safe.
LTS (Long-Term Support) 🟩	Companies / servers	Super stable version maintained for a long time. Only gets security fixes.

Capacity Limitations



- **Limitations on cloud capacity** can come from different sources:
 - **CSP limits and throttles:**
Cloud Service Providers (like AWS or Azure) limit how many resources you can use to control system load.
 - **Account owner caps:**
The account owner can set spending or usage limits.
 - AWS has four budget types: **cost**, **usage**, **savings plans**, and **reservation** budgets.
 - **Technology limitations:**
Sometimes limits come from hardware, software, or network performance.

In short: Cloud systems are not unlimited — they have rules and limits set by the provider, the account, and the technology itself.

Capacity Planning !!

- **Capacity planning** helps you organize and balance cloud resources effectively.
- Use the following techniques:
 - **Look to your SLAs:**
Check your Service Level Agreements to **understand resource and performance guarantees**.
 - **Balance reserved and on-demand services:**
Combine stable reserved services with flexible on-demand options for best performance.
 - **Establish utilization baselines:**
Create standard usage levels to guide future resource allocation and track performance trends.

In short: Good planning uses SLAs, balances service types, and monitors resource usage over time.

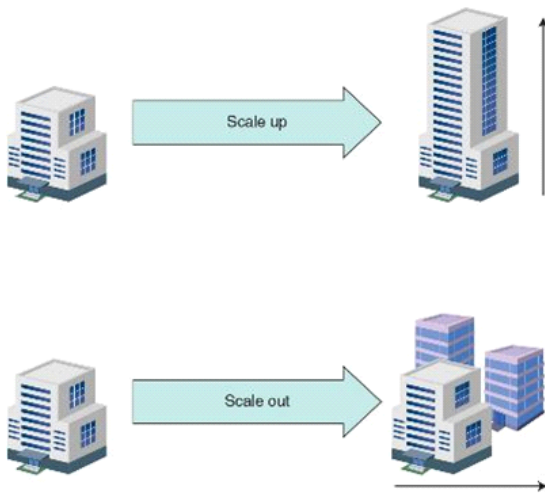


Figure 3-14 Scaling up increases the size of an existing resource while scaling out increases the number of resource instances

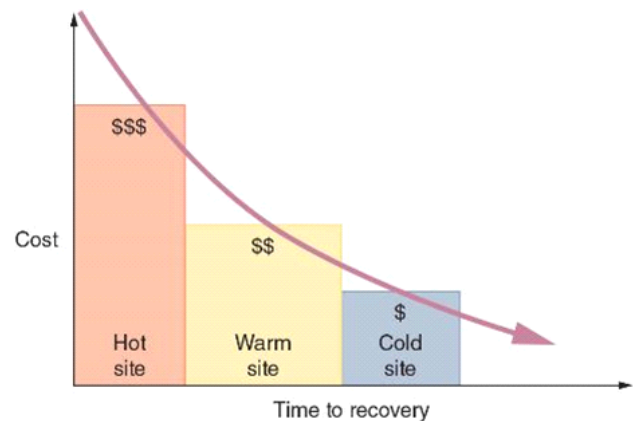


Figure 3-15 The most expensive option also provides the fastest recovery

Business Continuity Planning !!

- **Business continuity (BC)** ensures a company can keep working during a crisis, disaster or failure.
- A **Business Continuity Plan (BCP)** includes:
 - Contact info for emergency coordinators
 - Backup details (which data, how often, where stored, full recovery steps)
 - Network and redundancy details with service providers
 - Regular BCP testing strategies
 - Crisis management plan

Goal: Maintain operations and data safety during any disruption.

Organizations employ many different strategies as part of their BCP, such as the following:

- Defining **Roles and responsibilities**
- Creating **call trees** for communication
- Listing **Incident types and categories**
- Providing **training** for staff
- Using **Backup services**

Disaster Recovery!!

- **Disaster Recovery (DR)** is the process of returning systems and operations to normal after a crisis.
- Two key measures:
 - **RPO (Recovery Point Objective):** The maximum acceptable amount of data loss
 - **RTO (Recovery Time Objective):** The maximum acceptable downtime for a system to be restored after a failure.
- In cloud systems, **RPOs and RTOs** can often be reduced to nearly zero due to automation and replication.
- **Goal:** Recover data and resume full operations quickly after disruption.

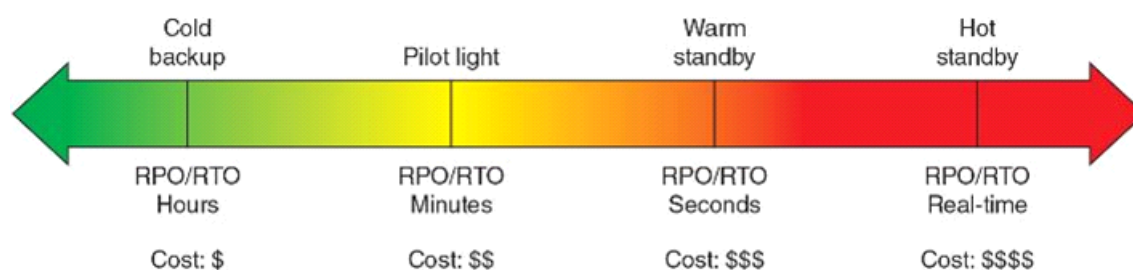


Figure 3-17 Various tiers of recovery services

This slide shows **different levels (tiers)** of disaster recovery — from slow and cheap to fast and expensive

- A **DR kit (Disaster Recovery Kit)** includes:
 - **BCP records** to restore access and backups
 - **Tools** to fix common or critical problems
 - **Emergency kits** for employee safety
 - **Incident response phases:**
 1. **Identification** – Find out what happened
 2. **Investigation** – Understand why it happened
 3. **Containment, eradication, and recovery** – Stop the issue and restore systems
 4. **Post-incident and lessons learned** – Review and improve for the future
- In short:* DR kits and clear response steps help organizations recover fast and prevent future issues.